

CH13.2 空間中的空間直線方程式

$$1.5x - 6y - 3z = 2 \quad 2. (1)4 \quad (2)\left(\frac{7}{5}, \frac{6}{5}, \frac{9}{5}\right)$$

$$3. 11, (26, 30, 36) \quad 4. \frac{x-1}{8} = \frac{y}{7} = \frac{z}{11} \quad 5. (A)(E)$$

$$6.4 \quad 7. \frac{5}{4} \quad 8. (-2, 4), \left(-\frac{1}{2}, \frac{1}{2}, 2\right)$$

$$9. (1) \left(\frac{3}{2}, 1, \frac{3}{2}\right) \quad (2) \sqrt{14}$$

$$10. (1) (3, 8, 3), (-3, -7, 6)$$

$$(2) \frac{x-3}{2} = \frac{y-8}{5} = \frac{z-3}{-1} \quad \text{或} \quad \frac{x+3}{2} = \frac{y+7}{5} = \frac{z-6}{-1}$$

解析

$$1. A(1, -1, 3) \in L_1, B(-2, 0, -4) \in L_2$$

$$\vec{v} = (3, 2, 1) \perp \vec{n} \quad \text{且} \quad \overrightarrow{AB} = (-3, 1, -7) \perp \vec{n}$$

$$\begin{aligned} \therefore \vec{n} &= \left(\begin{vmatrix} 2 & 1 \\ 1 & -7 \end{vmatrix}, \begin{vmatrix} 1 & 3 \\ -7 & -3 \end{vmatrix}, \begin{vmatrix} 3 & 2 \\ -3 & 1 \end{vmatrix} \right) \\ &= (-15, 18, 9) // (5, -6, -3) \end{aligned}$$

$$\text{令 } E: 5x - 6y - 3z = k, A(1, -1, 3) \text{ 代入}$$

$$\Rightarrow 5 + 6 - 9 = 2 = k \quad \therefore E: 5x - 6y - 3z = 2$$

$$2. (1) \text{ 設 } L_1 \text{ 上的點為 } (1 + 2t, 2 - kt, 1 + kt)$$

$$L_2 \text{ 上的點為 } \left(2 + 3s, 2 + ks, \frac{4}{5} - 5s\right)$$

$$\therefore \text{有交點} \therefore \text{方程組} \begin{cases} 1 + 2t = 2 + 3s \\ 2 - kt = 2 + ks \\ 1 + kt = \frac{4}{5} - 5s \end{cases} \text{有解}$$

$$\Leftrightarrow t = \frac{1}{5}, s = -\frac{1}{5}, k = \frac{4}{-}$$

(2) 將 $t = \frac{1}{5}, k = 4$ 代入 $(1 + 2t, 2 - kt, 1 + kt)$ 得交點坐標為 $(\frac{7}{5}, \frac{6}{5}, \frac{9}{5})$

3. (1) $\overrightarrow{AB} = (50, 60, 70) // (5, 6, 7)$

$$\therefore \overrightarrow{AB}: \begin{cases} x = 1 + 5t \\ y = 6t \\ z = 1 + 7t \end{cases}, 0 \leq t \leq 10$$

當 $t = 0, 1, 2, \dots, 9, 10$, 可得 11 個格子點

(2) $d = \frac{|2+10t+12t+1+7t-150|}{\sqrt{4+4+1}} = \frac{|29t-147|}{3}$

$$t = 0, 1, 2, \dots, 9, 10$$

當 $t = 5, d$ 會有最小值, 此時 $(x, y, z) = (26, 30, 36)$

4. 設點 A 在直線 L 的垂足為 $P(1 + t, -1 + 2t, 3 - 2t)$

則 $\overrightarrow{AP} = (t, -1 + 2t, 3 - 2t)$

由 $\overrightarrow{AP} \perp \vec{\ell}$ 知: $(t, -1 + 2t, 3 - 2t) \cdot (1, 2, -2) = 0 \Rightarrow 9t - 8 = 0 \Rightarrow t = \frac{8}{9}$

$\therefore \overrightarrow{AP} = (\frac{8}{9}, \frac{7}{9}, \frac{11}{9}) // (8, 7, 11)$ 即垂直線的方向向量

故方程式為 $\frac{x-1}{8} = \frac{y}{7} = \frac{z}{11}$

5. 直線與平面沒有交點 $\Rightarrow$ 直線與平面平行 $\Rightarrow$ 直線方向向量與平面法向量垂直, 且直線不在平面上

(A) $(3, -1, 2) \cdot (2, 4, -1) = 6 - 4 - 2 = 0$

且點 $(1, -2, 2)$ 不在平面上, 直線與平面平行

$\Rightarrow$ 沒有交點

(B) $(3, -1, 2) \cdot (3, 1, 2) = 9 - 1 + 4 = 12 \Rightarrow$ 直線與平面不平行 $\Rightarrow$ 有交點

(C) $(3, -1, 2) \cdot (3, 1, -4) = 9 - 1 - 8 = 0$,

但直線上的點 $(2, 5, 2)$ 在平面上

$\Rightarrow$ 整條直線落在該平面上 $\Rightarrow$ 有無限多個交點

(D) 將 $x = y = 0$ 代入 $E: 3x - y + 2z = 5$ 得 $z = \frac{5}{2} \Rightarrow$ 交點為 $(0, 0, \frac{5}{2})$

(E) $(3, -1, 2) \cdot (1, 1, -1) = 3 - 1 - 2 = 0$ 且點

$(0, -1, 3)$ 不在平面上

$\Rightarrow$ 直線與平面平行 $\Rightarrow$ 沒有交點

故選(A)

$$6. \vec{v}_L = (1, -1, 1) \times (2, -1, -1)$$

$$= \left(\begin{vmatrix} -1 & 1 \\ -1 & -1 \end{vmatrix}, \begin{vmatrix} 1 & 1 \\ -1 & 2 \end{vmatrix}, \begin{vmatrix} 1 & -1 \\ 2 & -1 \end{vmatrix} \right)$$

$$= (2, 3, 1)$$

$$\because L // E$$

$$\therefore \vec{v}_L \perp \vec{n}_E \Rightarrow (2, 3, 1) \cdot (a, -3, 1) = 0 \Rightarrow a = \underline{4}$$

$$7. \text{兩直線之交點，即為方程組} \begin{cases} x - y - z = 1 \\ x + y + z = 0 \\ 2x + 3y + z = 3 \end{cases} \text{之解} \Rightarrow x = \frac{1}{2}, y = \frac{5}{4}, z = -\frac{7}{4}$$

$$\text{此交點亦滿足 } x + 2y + z = a \Rightarrow a = \frac{1}{2} + \frac{5}{2} + \left(-\frac{7}{4}\right) = \frac{5}{4}$$

$$8. \because L \perp E$$

$$\therefore L \text{ 的方向向量 } \vec{d} = (6, b, c) //$$

$$E \text{ 的法向量 } \vec{n} = (3, -1, 2)$$

$$(1) \frac{6}{3} = \frac{b}{-1} = \frac{c}{2}, \text{得 } b = -2, c = 4$$

$$\text{故數對 } (b, c) = (-2, 4)$$

$$(2) \because H \in L: \begin{cases} x = -2 + 6t \\ y = 1 - 2t \\ z = 1 + 4t \end{cases}$$

$$\therefore \text{設 } H(-2 + 6t, 1 - 2t, 1 + 4t)$$

$$\because H \in E \therefore H \text{ 代入 } E$$

$$\Rightarrow (-6 + 18t) - (1 - 2t) + (2 + 8t) = 2 \Rightarrow t = \frac{1}{4}$$

則 $H\left(-\frac{1}{2}, \frac{1}{2}, 2\right)$

9. $A(1, \sqrt{3}), B(3, 0, 1)$

$$L: \frac{x-1}{2} = \frac{y}{4} = \frac{z-2}{-2} \Rightarrow x-1 = \frac{y}{2} = \frac{z-2}{-1} = t$$

L 上任一點 $P(t+1, 2t, -t+2)$

$$\begin{aligned} & \overline{PA} + \overline{PB} \\ &= \sqrt{t^2 + (2t-2)^2 + (-1-t)^2} + \sqrt{(t-2)^2 + (2t)^2 + (1-t)^2} \\ &= \sqrt{6t^2 - 6t + 5} + \sqrt{6t^2 - 6t + 5} \\ &= 2\sqrt{6\left(t - \frac{1}{2}\right)^2 + \frac{7}{2}} \end{aligned}$$

當 $t = \frac{1}{2}$ 時, $\overrightarrow{PA} + \overrightarrow{PB}$ 最小值 $= 2\sqrt{\frac{7}{2}} = \sqrt{14}$

此時 P 的坐標為 $\left(\frac{3}{2}, 1, \frac{3}{2}\right)$

10. $L_1: \frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}, L_2: \frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$

(1) 設公垂線 L_1 , L_2 上之垂足分別為 $P(3+3s, 8-s,$

$3+s)$ 與 $Q(-3-3t, -7+2t, 6+4t)$

則 $\overrightarrow{PQ} = (-3s-3t-6, s+2t-15, -s+4t+3)$

又直線 L_1 與 L_2 之方向向量各為 $\vec{\ell} = (3, -1, 1)$,

$$\vec{\ell}_2 = (-3, 2, 4)$$

由 $\overrightarrow{PQ} \perp L_1$

$$\Rightarrow \overrightarrow{PQ} \cdot \vec{\ell}_1 = 3(-3s-3t-6) - (s+2t-15) + (-s+4t+3)$$

$$= -11s - 7t = 0 \Rightarrow 11s + 7t = 0 \dots \textcircled{1}$$

由 $\overrightarrow{PQ} \perp L_2$

$$\Rightarrow \overrightarrow{PQ} \cdot \overrightarrow{\ell_2} = -3(-3s - 3t - 6) + 2(s + 2t - 15) +$$

$$4(-s + 4t + 3) = 7s + 29t = 0 \Rightarrow 7s + 29t = 0 \cdots \textcircled{2}$$

解 $\textcircled{1}\textcircled{2}$ 得 $s = t = 0$

故 $P(3, 8, 3), Q(-3, -7, 6)$

$$(2) \because \overrightarrow{PQ} = (-6, -15, 3) = -3(2, 5, -1)$$

$\therefore (2, 5, -1)$ 為公垂線 $\overrightarrow{PQ}$ 的方向向量故直線 PQ 的方程式為

$$\frac{x-3}{2} = \frac{y-8}{5} = \frac{z-3}{-1} \text{ 或 } \frac{x+3}{2} = \frac{y+7}{5} = \frac{z-6}{-1}$$